

The ANNNI phase diagram under two classes of error

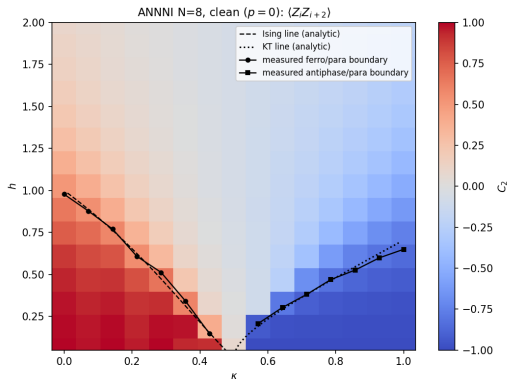
What depolarizing noise and coherent drift each do to a phase boundary, and what recovers it

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Depolarizing noise fades a phase diagram. Coherent drift moves it. Different mitigations fix each, and together they separate what calibration can recover from what decoherence costs.

1. The model, the observable, and the clean diagram



$H = -\sum Z_i Z_{i+1} + \kappa \sum Z_i Z_{i+2} - h \sum X_i$, $N = 8$,
periodic, 15×15 grid.

Observable: $C_2 = \frac{1}{N} \sum \langle Z_i Z_{i+2} \rangle$. +1 ferromagnet,
-1 antiphase ($\uparrow\uparrow\downarrow\downarrow$), ≈ 0 paramagnet.

Symmetry-safe at finite size.

Thresholds fixed once on the analytic lines and
never changed.

Ferro/para at $\kappa = 0.29$: $h^* = 0.512$ (Ising line
0.473).

Antiphase/para at $\kappa = 0.71$: $h^* = 0.381$ (KT line
0.381).

$\kappa = 0.5$: multicritical, no ordered reference,
handled separately.

2. Preparation and the two error classes

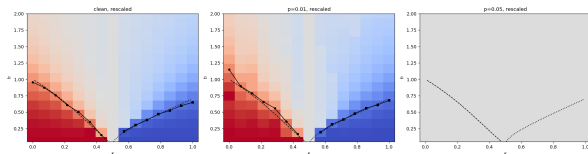
Circuit. Hamiltonian-variational ansatz from $|+\rangle^{\otimes N}$: four layers of $e^{-ig_1 \sum ZZ_{nn}} e^{-ig_2 \sum ZZ_{nnn}} e^{-ib \sum X}$, 12 parameters, 128 CNOTs. L-BFGS with adjoint gradients, warm-started from the paramagnet. Median energy error 4×10^{-5} ; C_2 within 0.05 of exact diagonalization at 95% of points.

Error class 1, the track's question. `DepolarizingChannel( $p$ )` on the target after every CNOT, $p = 0.01$ and 0.05 , on PennyLane `default.mixed`, all 225 points.

Error class 2, what every processor also has. Every RX angle $\times(1 + \varepsilon_i)$, every RZZ angle $\times(1 + \delta_k)$, drifting across the scan: static offset plus wander, common-mode plus independent, RMS 3% and 6%.

Two mitigations. Rescale by a reference point (measure the decay). Or hold the angles in-band: five probe circuits per cycle feeding a per-channel controller that corrects only above the noise floor and learns each slope from its own motion.

3. Depolarizing is amplitude loss



Deep-ordered signal: $0.92 \rightarrow 0.48$ at $p = 0.01, 0.05$ at $p = 0.05$.

Fixed threshold: every ordered point lost at both levels (24.9%).

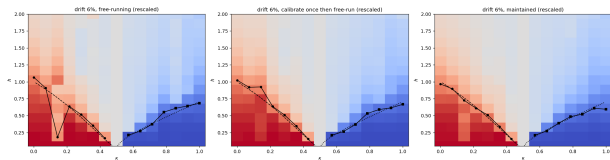
Rescaled by the $h = 0.05$ reference:

$p = 0.01$ recovers to 1.3% misclassified; ferro/para $0.504 \rightarrow 0.559$, antiphase/para $0.381 \rightarrow 0.388$. $p = 0.05$: nothing recovers it.

Most robust: the antiphase (0.60 of clean vs 0.37 for the ferromagnet at $p = 0.01$). The ferromagnet is built with more X rotation late in the circuit, after the CNOTs that carry the noise. The opposite of the naive guess.

What matters is $p \times \text{depth}$: 256 CNOTs lose the phases at $p = 0.01$ outright.

4. Coherent drift is shape distortion



Signed and first order: +10% over-rotation moves C_2 by -0.056 , under-rotation the other way. Boundaries move; rescaling cannot help.

6% drift, free-running: mean shift 0.075, one excursion of 0.58, 4.4% misclassified.

Calibrate once, then free-run: 0.037 (the static half).

Maintained in-band: 0.015, 1.8% misclassified, 80% of the shift removed.
At 3%: 0.016 $\rightarrow$ 0.007.

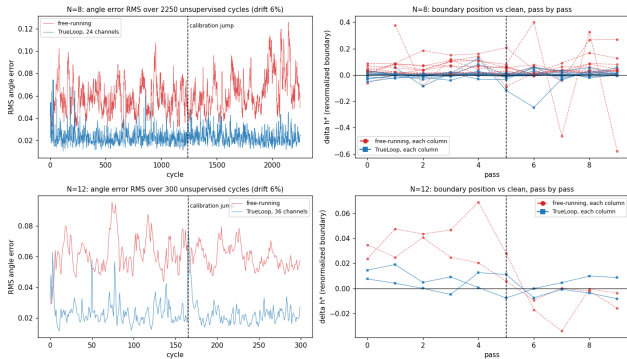
The ferromagnetic boundary moves $3\times$ more than the antiphase boundary: C_2 is steeper there.

5. What recovers what, and at what cost

Arm (rescaled boundaries, $N = 8$)	misclassified	mean $ \Delta h^* $	operations
depolarizing $p = 0.01$	1.3%	0.035	rescaling, no circuits
depolarizing $p = 0.05$	24.9%	none	unrecoverable
drift 6%, free-running	4.4%	0.075	none
drift 6%, calibrate once at the start	1.3%	0.037	one sweep
drift 6%, re-optimize the VQE at every point	0.0% (sub-grid)		about 36,000 circuits per point
drift 6%, maintained in-band	1.8%	0.015	5 probe circuits per cycle
drift 6% and $p = 0.01$ together, free / maintained, measured against the depolarizing-only diagram	3.6% / 0.4%	0.070 / 0.038	boundaries land on the depolarizing-only ones

Under both errors, maintenance returns the diagram to its depolarizing-only form: the incoherent part is isolated, and it is the only part left.

6. Continuous, unsupervised, in real time



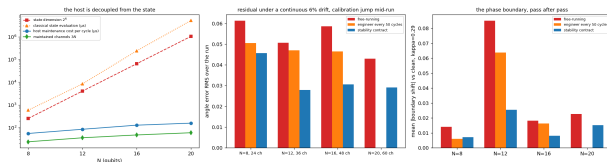
Ten passes, 2,250 cycles, 6% drift, every offset redrawn at pass 5, nobody intervening.

$N = 8$, 24 channels: residual 0.062 $\rightarrow$ 0.024 every pass; boundary std 0.051 $\rightarrow$ 0.020; jump absorbed in 3 cycles; host 47 μ s per cycle.

$N = 12$, 36 channels: 0.062 $\rightarrow$ 0.025; std 0.025 $\rightarrow$ 0.007; 76 μ s per cycle. Linear in channels.

Free-running walks away by the last pass. The maintained diagram does not move.

7. Complexity decoupled from the state



Two things keep continuous hardware in a lab: a person recalibrating it on a schedule, which cannot follow a wander faster than the schedule; or a host reconstructing its state, which costs n^2 memory and cannot reach high dimension. The contract does neither: $3N$ channels from one read, while the state has 2^N amplitudes.

$N = 8 \rightarrow 20$: state dimension $\times 4,096$;
 classical state evaluation $1 \text{ ms} \rightarrow 5.2 \text{ s}$; host
 maintenance cost $56 \rightarrow 159 \mu\text{s}$, linear at
 $2.6 \mu\text{s}$ per channel.

Three arms on the same scan: free-running, an engineer every 50 cycles, the contract. At every size the contract holds the boundary tighter than the scheduled engineer; at $N = 12$, three times tighter ($0.085 / 0.064 / 0.025$).

The host never touches the state, so its burden does not grow with it. That is what lets a hybrid system scale.

8. Interpretation, limitations, what is next

In the model's language. Depolarizing shortens every correlation length uniformly: order parameters shrink, boundaries stay, a ratio observable sees them until the correlation length is below two sites. Coherent drift changes the effective Hamiltonian the circuit prepares, shifting where frustration and field balance: the boundary really moves. Rescaling fixes one, maintenance fixes the other.

Not resolved. The floating phase, at $N = 8$; the two poorly converged points near $\kappa \approx 0.6$ sit exactly there.

Limitations. $N = 8$ periodic; noise after CNOTs only; the wander speed sets the maintenance floor at 0.4; the eight-layer circuits are exact at $N = 12$ and at their limit by $N = 16$. One product boundary found: with declared signs and a low-SNR probe the controller mis-identified channels and held them; letting it learn the signs fixed it.

Everything reproduces. Executed notebook, PennyLane grids, cached data, code, and the maintenance layer's evaluation build are in the package.